

Homework 1: When Do Estimates Miss?

POLS 602

Due: Oct. 2, 2026

Instructions

Submit both your code and answers by pushing them to your github repo. They may be in separate files, or in a single file like a .qmd, .rmd, or .ipynb notebook. Comment your code and make sure it executes without errors.

Monte Carlo Simulation

Real data give us a single sample and estimate, so we can never see how far that estimate is from the true value. A Monte Carlo simulation gets around this by simulating data. You choose a data-generating process with known parameter values, and then repeat the following many times:

1. Generate a new dataset from the process.
2. Apply the estimator to the dataset.
3. Record the estimate.

After many repetitions, the recorded estimates approximate the estimator's sampling distribution. Because you know the true parameter, you can compute the estimator's bias (the mean of the estimates minus the true value) and how much it varies from sample to sample (the standard deviation of the estimates).

Some functions that may be helpful for this assignment:

```
# Helpful functions
set.seed(seed)      # makes random draws reproducible
rnorm(n, mean, sd)  # draws n values from a normal distribution, e.g. to generate data
replicate(reps, expr) # runs expr reps times, e.g. to repeat a simulation
var(x); sd(x)       # sample variance and standard deviation of a vector
cor(x, y)           # correlation of two vectors; cor(df) gives a correlation matrix
lm(y ~ x, data = d) # fits a linear regression
coef(model)         # extracts the estimated coefficients from a fitted model
summary(model)      # coefficient table with standard errors
confint(model)      # confidence intervals for the coefficients
data.frame(a, b)     # combines vectors into a data frame
rbind(df1, df2)      # stacks data frames; do.call(rbind, list_of_dfs) stacks a list of them
sapply(x, fn)        # applies fn to each element of x and returns a vector
all.equal(a, b)      # checks whether two numbers are equal up to floating-point error
```

Problem 1: Why does the sample variance divide by $n - 1$?

The sample variance divides by $n - 1$ rather than n . In this problem you will use a Monte Carlo simulation to see why.

The population of our variable of interest is $N(0, 2)$ (normally distributed with a mean of 0 and standard deviation of 2), so the true variance is 4. You will compare three estimators computed from a sample:

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2}$$

- (a) Write a function that takes a numeric vector and returns the three estimates. R's `var()` and `sd()` compute the sample versions (dividing by $n - 1$), so you will need to implement the divide-by- n version yourself. Check your function on a simulated sample of 10 observations.
- (b) Simulate 1,000 samples of 5 observations from the population and apply your function to each. Store the results in a data frame with one row per sample. Plot the sampling distributions of the two variance estimators on the same axes, marking the true variance and each estimator's mean. Which estimator is biased, and in which direction?
- (c) Repeat (b) with samples of 3, 5, 10, 30, and 100 observations. Plot the estimated bias of each variance estimator against sample size.
- (d) Is s an unbiased estimator of $\sigma = 2$? Report its estimated bias at each sample size and explain what you find.
- (e) At $n = 5$, compare the standard deviations of the 1,000 estimates of each variance estimator. Which is less variable? Why doesn't that make it the better estimator?

Problem 2: Omitted Variable Bias

Suppose we want to estimate the effect of news consumption (x) on political knowledge (y). The true data generating process is:

$$\begin{aligned}z_i &\sim N(0, 1) \\x_i &= \rho z_i + \sqrt{1 - \rho^2} u_i, \quad u_i \sim N(0, 1) \\y_i &= \beta_0 + \beta_1 x_i + \beta_2 z_i + \epsilon_i, \quad \epsilon_i \sim N(0, 1)\end{aligned}$$

where z_i , u_i , and ϵ_i are independent, $\beta_0 = 1$, $\beta_1 = 2$, $\beta_2 = 3$, and ρ is the correlation between x_i and z_i . The estimand is $\beta_1 = 2$. You will compare the theorized model, $y \sim x$ (a researcher who does not account for political interest), with the true model, $y \sim x + z$.

- (a) Generate a dataset of 100 observations with $\rho = 0.5$. You may want to check that your variables have the distributions you expect.
- (b) Fit both models, plot y against x , and describe your findings.
- (c) Repeat (a) and (b) with 1,000 observations. Did more data resolve the bias?
- (d) Repeat (c) with $\rho = 0$. Explain your findings. What does this imply for how we should construct regression models?